

# 05

## Continuity

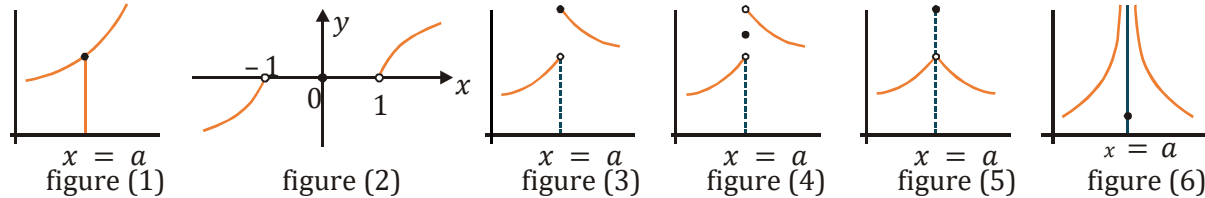
### 1. CONTINUOUS FUNCTIONS:

A function for which a small change in the independent variable causes only a small change and not a sudden jump in the dependent variable are called continuous functions. Naively, we may say that a function is continuous at a fixed point if we can draw the graph of the function around that point without lifting the pen from the plane of the paper.

A function  $f(x)$  is said to be continuous at  $x = a$ , if  $\lim_{x \rightarrow a} f(x)$  exists and is equal to  $f(a)$ . Symbolically

$f(x)$  is continuous at  $x = a$  if  $\lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+h) = f(a) = \text{finite quantity}$ .

i.e.  $\text{LHL}|_{x=a} = \text{RHL}|_{x=a} = \text{value of } f(x)|_{x=a} = \text{finite quantity. } (h > 0)$



In figure (1) and (2)  $f(x)$  is continuous at  $x = a$  and  $x = 0$  respectively and in figure (3) to (6)  $f(x)$  is discontinuous at  $x = a$ .

**Note 1 :** Continuity of a function must be discussed only at points which are in the domain of the function.

**Note 2 :** If  $x = a$  is an isolated point of domain then  $f(x)$  is always considered to be continuous at  $x = a$ .

#### Illustration 1

If  $f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x], & x \geq 1 \end{cases}$  then find whether  $f(x)$  is continuous or not at  $x = 1$ , where  $[.]$  denotes greatest integer function.

integer function.

**Solution:**

$$f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x], & x \geq 1 \end{cases}$$

For continuity at  $x = 1$ , we determine,  $f(1)$ ,  $\lim_{x \rightarrow 1^-} f(x)$  and  $\lim_{x \rightarrow 1^+} f(x)$ .

Now,  $f(1) = [1] = 1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \sin \frac{\pi x}{2} = \sin \frac{\pi}{2} = 1 \text{ and } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} [x] = 1$$

so  $f(1) = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$

$\therefore f(x)$  is continuous at  $x = 1$

**Illustration 2**

$$\text{Let } f(x) = \begin{cases} \frac{a(1-x\sin x) + b\cos x + 5}{x^2} & x < 0 \\ 3 & x = 0 \\ \left(1 + \left(\frac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} & x > 0 \end{cases}$$

If  $f$  is continuous at  $x = 0$ , then find out the values of  $a, b, c$  and  $d$ .

**Solution:**

Since  $f(x)$  is continuous at  $x = 0$ , so at  $x = 0$ , both left and right limits must exist and both must be equal to 3.

$$\text{Now } \lim_{x \rightarrow 0^-} \frac{a(1-x\sin x) + b\cos x + 5}{x^2} = \lim_{x \rightarrow 0^-} \frac{(a+b+5) + \left(-a - \frac{b}{2}\right)x^2 + \dots}{x^2} = 3 \quad (\text{By the expansions of } \sin x \text{ and } \cos x)$$

$$\text{If } \lim_{x \rightarrow 0^-} f(x) \text{ exists then } a + b + 5 = 0 \text{ and } -a - \frac{b}{2} = 3 \Rightarrow a = -1 \text{ and } b = -4$$

$$\text{since } \lim_{x \rightarrow 0^+} \left(1 + \left(\frac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} \text{ exists } \Rightarrow \lim_{x \rightarrow 0^+} \frac{cx + dx^3}{x^2} = 0 \Rightarrow c = 0$$

$$\text{Now } \lim_{x \rightarrow 0^+} (1 + dx)^{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \left[(1 + dx)^{\frac{1}{dx}}\right]^d = e^d$$

$$\text{So } e^d = 3 \Rightarrow d = \ln 3,$$

$$\text{Hence } a = -1, b = -4, c = 0 \text{ and } d = \ln 3.$$

**2. CONTINUITY OF THE FUNCTION IN AN INTERVAL:**

- (a) A function is said to be continuous in  $(a, b)$  if  $f$  is continuous at each & every point belonging to  $(a, b)$ .
- (b) A function is said to be continuous in a closed interval  $[a, b]$  if :
  - (i)  $f$  is continuous in the open interval  $(a, b)$
  - (ii)  $f$  is right continuous at ' $a$ ' i.e.  $\lim_{x \rightarrow a^+} f(x) = f(a) = \text{a finite quantity}$
  - (iii)  $f$  is left continuous at ' $b$ ' i.e.  $\lim_{x \rightarrow b^-} f(x) = f(b) = \text{a finite quantity}$

**Note:**

- (i) All polynomials, trigonometrical functions, exponential & logarithmic functions are continuous in their domains.
- (ii) If  $f(x)$  &  $g(x)$  are two functions that are continuous at  $x = c$  then the function defined by :  
 $F_1(x) = f(x) \pm g(x)$ ;  $F_2(x) = K f(x)$ , where  $K$  is any real number  $F_3(x) = f(x) \cdot g(x)$ ; are also continuous at  $x = c$ .

$$\text{Further, if } g(c) \text{ is not zero, then } F_4(x) = \frac{f(x)}{g(x)} \text{ is also continuous at } x = c.$$

## Illustration 3

If  $f(x) = \begin{cases} \frac{\sin(p+1) + \sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} & , x > 0 \end{cases}$  is continuous at  $x = 0$ , then the ordered pair  $(p, q)$  is equal to:

- (A)  $\left(\frac{5}{2}, \frac{1}{2}\right)$  (B)  $\left(-\frac{3}{2}, -\frac{1}{2}\right)$  (C)  $\left(-\frac{1}{2}, \frac{3}{2}\right)$  (D)  $\left(-\frac{3}{2}, \frac{1}{2}\right)$

Ans. (D)

Solution:

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} = \lim_{x \rightarrow 0^+} \frac{\sqrt{1+x} - 1}{x} = \frac{1}{2}$$

$$\text{LHL} = \lim_{x \rightarrow 0} \frac{\sin(p+1)x + \sin x}{x} = (p+1) + 1 = p+2$$

for continuity  $\text{LHL} = \text{RHL} = f(0)$

$$\Rightarrow (p, q) = \left(-\frac{3}{2}, \frac{1}{2}\right)$$

## Illustration 4

If the function  $f$  defined on  $\left(-\frac{1}{3}, \frac{1}{3}\right)$  by  $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+3x}{1-2x}\right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$  is continuous, then  $k$  is equal

to \_\_\_\_\_

Ans. (5.00)

Solution:

$$k = \lim_{x \rightarrow 0} \left( \frac{\ln(1+3x)}{x} - \frac{\ln(1-2x)}{x} \right)$$

$$k = 3 + 2 = 5$$

## Illustration 5

If  $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} & ; x < 0 \\ b & ; x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}} & ; x > 0 \end{cases}$  is continuous at  $x = 0$ , then  $a + 2b$  is equal to :

- (A) -1 (B) 1 (C) -2 (D) 0

Ans. (D)

**Solution:**

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left( \frac{\sin(a+2)x}{x} + \frac{\sin x}{x} \right) = a+3$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}}$$

$$= \lim_{x \rightarrow 0^-} \frac{(1+3x)^{1/3} - 1}{x} = 1$$

$$f(0) = b$$

for continuity at  $x = 0$

$$\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$$

$$\Rightarrow a+3 = b = 1$$

$$\therefore a = -2, b = 1$$

$$\therefore a+2b = 0$$

**Illustration 6**

Let  $f : R \rightarrow R$  be a function defined as  $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin 2x}{2x} & , \text{ if } x < 0 \\ b & , \text{ if } x = 0 \\ \frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}} & , \text{ if } x > 0 \end{cases}$ . If  $f$  is continuous at  $x = 0$ ,

then the value of  $a+b$  is equal to :

(A)  $-\frac{5}{2}$

(B)  $-2$

(C)  $-3$

(D)  $-\frac{3}{2}$

**Ans. (D)**

**Solution:**

$f(x)$  is continuous at  $x = 0$

$$\lim_{x \rightarrow 0^+} f(x) = f(0) = \lim_{x \rightarrow 0^-} f(x) \quad \dots(1)$$

$$f(0) = b \quad \dots(2)$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \left( \frac{\sin(a+1)x}{2x} + \frac{\sin 2x}{2x} \right) \\ &= \frac{a+1}{2} + 1 \quad \dots(3) \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}} \\ &= \lim_{x \rightarrow 0^+} \frac{(x+bx^3-x)}{bx^{5/2}(\sqrt{x+bx^3} + \sqrt{x})} \\ &= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{x}(\sqrt{1+bx^2} + 1)} = \frac{1}{2} \quad \dots(4) \end{aligned}$$

Use (2), (3) & (4) in (1)

## Continuity

$$\frac{1}{2} = b = \frac{a+1}{2} + 1$$

$$\Rightarrow b = \frac{1}{2}, a = -2$$

$$a + b = \frac{-3}{2}$$

### Illustration 7

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined as  $f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} \log_e \left( \frac{1 + 2xe^{-2x}}{(1 - xe^{-x})^2} \right), & x \neq 0 \\ \alpha, & x = 0 \end{cases}$ . If  $f$  is continuous at  $x = 0$ ,

then  $\alpha$  is equal to :

(A) 1

(B) 3

(C) 0

(D) 2

**Ans. (A)**

**Solution:**

For continuity

$$\lim_{x \rightarrow 0} \frac{x^3}{4 \sin^4 x} (\ln(1 + 2xe^{-2x}) - 2 \ln(1 - xe^{-x}))$$

$$= \alpha$$

$$\lim_{x \rightarrow 0} \frac{1}{4x} [2xe^{-2x} + 2xe^{-x}] = \alpha$$

$$= \frac{1}{4}(4) = \alpha = 1$$

### Illustration 8

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined as  $f(x) = \begin{cases} \frac{\lambda |x^2 - 5x + 6|}{\mu(5x - x^2 - 6)}, & x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}}, & x > 2 \\ \mu, & x = 2 \end{cases}$ , where  $[x]$  is the greatest integer less than or

equal to  $x$ . If  $f$  is continuous at  $x = 2$ , then  $\lambda + \mu$  is equal to :

(A)  $e(-e + 1)$

(B)  $e(e - 2)$

(C) 1

(D)  $2e - 1$

**Ans. (A)**

**Solution:**

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} e^{\frac{\tan(x-2)}{x-2}} = e^1$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{-\lambda(x-2)(x-3)}{\mu(x-2)(x-3)} = -\frac{\lambda}{\mu}$$

$$\text{For continuity } \mu = e = -\frac{\lambda}{\mu} \Rightarrow \mu = e, \lambda = -e^2$$

$$\lambda + \mu = e(-e + 1)$$

**Illustration 9**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined as  $f(x) = \begin{cases} 2\sin\left(-\frac{\pi x}{2}\right), & \text{if } x < -1 \\ |ax^2 + x + b|, & \text{if } -1 \leq x \leq 1 \\ \sin(\pi x), & \text{if } x > 1 \end{cases}$ . If  $f(x)$  is continuous on  $\mathbb{R}$ , then  $a + b$

equals:

(A) -3

(B) -1

(C) 3

(D) 1

**Ans. (B)**

**Solution:**

$f(x)$  is continuous on  $\mathbb{R}$

$$\Rightarrow f(1^-) = f(1) = f(1^+)$$

$$|a + 1 + b| = \lim_{x \rightarrow 1} \sin(\pi x)$$

$$|a + 1 + b| = 0 \Rightarrow a + b = -1 \quad \dots(1)$$

$$\Rightarrow \text{Also } f(-1^-) = f(-1) = f(-1^+)$$

$$\lim_{x \rightarrow -1} 2\sin\left(-\frac{\pi x}{2}\right) = |a - 1 + b|$$

$$|a - 1 + b| = 2$$

$$\text{Either } a - 1 + b = 2 \text{ or } a - 1 + b = -2$$

$$a + b = 3 \quad \dots(2)$$

$$\text{or } a + b = -1 \quad \dots(3)$$

$$\text{from (1) and (2)} \Rightarrow a + b = 3 = -1 \text{ (reject)}$$

$$\text{from (1) and (3)} \Rightarrow a + b = -1$$

**Illustration 10**

Let  $a, b \in \mathbb{R}, b \neq 0$ , Define a function  $f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1), & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & \text{for } x > 0. \end{cases}$ . If  $f$  is continuous at  $x = 0$ , then

$10 - ab$  is equal to \_\_\_\_\_.

**Ans. (14)**

**Solution:**

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1), & x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & x > 0 \end{cases}$$

For continuity at '0'

$$\lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\tan 2x - \sin 2x}{bx^3} = -a$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\frac{8x^3}{3} + \frac{8x^3}{3!}}{bx^3} = -a$$

$$\Rightarrow 8\left(\frac{1}{3} + \frac{1}{3!}\right) = -ab$$

$$\Rightarrow 4 = -ab$$

$$\Rightarrow 10 - ab = 14$$

**Illustration 11**

Let  $f : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow \mathbb{R}$  be defined as  $f(x) = \begin{cases} (1 + |\sin x|)^{\frac{3a}{|\sin x|}}, & -\frac{\pi}{4} < x < 0 \\ b, & x = 0 \\ e^{\cot 4x / \cot 2x}, & 0 < x < \frac{\pi}{4} \end{cases}$ .

If  $f$  is continuous at  $x = 0$ , then the value of  $6a + b^2$  is equal to:

(A)  $1 - e$

(B)  $e - 1$

(C)  $1 + e$

(D)  $e$

**Ans. (C)**

**Solution:**

$$\lim_{x \rightarrow 0} f(x) = b$$

$$\lim_{x \rightarrow 0^+} x e^{\frac{\cot 4x}{\cot 2x}} = e^{\frac{1}{2}} = b$$

$$\lim_{x \rightarrow 0^-} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} = e^{3a} = e^{\frac{1}{2}}$$

$$\lim_{x \rightarrow 0^-} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} = e^{3a} = e^{\frac{1}{2}}$$

$$a = \frac{1}{6} \Rightarrow 6a = 1$$

$$(6a + b^2) = (1 + e)$$

**3. THE INTERMEDIATE VALUE THEOREM:**

Suppose  $f(x)$  is continuous on an interval  $I$ , and  $a$  and  $b$  are any two points of  $I$ . Then if  $y_0$  is a number between  $f(a)$  and  $f(b)$ , there exists a number  $c$  between  $a$  and  $b$  such that  $f(c) = y_0$ .

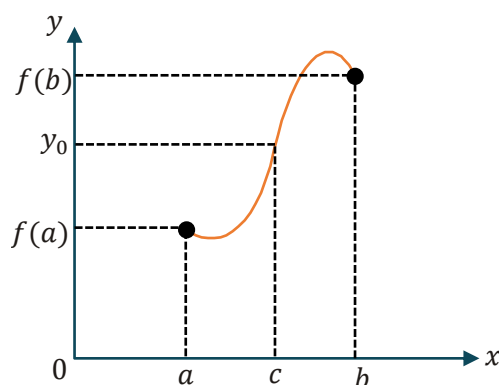
Note that a function  $f$  which is continuous in  $[a, b]$  possesses the following properties:

- (i) If  $f(a)$  &  $f(b)$  possess opposite signs, then there exists at least one root of the equation  $f(x) = 0$  in the open interval  $(a, b)$ .
- (ii) If  $K$  is any real number between  $f(a)$  &  $f(b)$ , then there exists at least one root of the equation  $f(x) = K$  in the open interval  $(a, b)$ .

**Note:** In above cases the number of roots is always odd.

**Illustration 12**

Show that the function,  $f(x) = (x - a)^2(x - b)^2 + x$ , takes the value  $\frac{a+b}{2}$  for some  $x_0 \in (a, b)$



The function  $f$ , being continuous on  $[a, b]$  takes on every value between  $f(a)$  and  $f(b)$

**Solution:**

$$f(x) = (x-a)^2(x-b)^2 + x$$

$$f(a) = a$$

$$f(b) = b$$

$$\& \frac{a+b}{2} \in (f(a), f(b))$$

$\therefore$  By intermediate value theorem, there is atleast one  $x_0 \in (a, b)$  such that  $f(x_0) = \frac{a+b}{2}$ .

**Illustration 13**

Let  $f: [0, 1] \xrightarrow{\text{onto}} [0, 1]$  be a continuous function, then prove that  $f(x) = x$  for atleast one  $x \in [0, 1]$

**Solution:**

Consider  $g(x) = f(x) - x$

$$g(0) = f(0) - 0 = f(0) \geq 0 \quad \{\because 0 \leq f(x) \leq 1\}$$

$$g(1) = f(1) - 1 \leq 0$$

$$\Rightarrow g(0) \cdot g(1) \leq 0$$

$$\Rightarrow g(x) = 0 \text{ has atleast one root in } [0, 1]$$

$$\Rightarrow f(x) = x \text{ for atleast one } x \in [0, 1]$$

**4. SOME IMPORTANT POINTS:**

- (a) If  $f(x)$  is continuous &  $g(x)$  is discontinuous at  $x = a$  then the product function  $\phi(x) = f(x) \cdot g(x)$  will **not necessarily be discontinuous at  $x = a$** , e.g.

$$f(x) = x \& g(x) = \begin{cases} \sin \frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$  is continuous at  $x = 0$  &  $g(x)$  is discontinuous at  $x = 0$ , but  $f(x) \cdot g(x)$  is continuous at  $x = 0$ .

- (b) If  $f(x)$  and  $g(x)$  both are discontinuous at  $x = a$  then the product function  $\phi(x) = f(x) \cdot g(x)$  **is not necessarily be discontinuous at  $x = a$** , e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

$f(x)$  &  $g(x)$  both are discontinuous at  $x = 0$  but the product function  $f \cdot g(x)$  is still continuous at  $x = 0$

- (c) If  $f(x)$  and  $g(x)$  both are discontinuous at  $x = a$  then  $f(x) \pm g(x)$  is not necessarily be discontinuous at  $x = a$

- (d) A continuous function whose domain is closed must have a range also in closed interval.

- (e) If  $f$  is continuous at  $x = a$  &  $g$  is continuous at  $x = f(a)$  then the composite  $g[f(x)]$  is continuous

at  $x = a$ . eg.  $f(x) = \frac{x \sin x}{x^2 + 2}$  &  $g(x) = |x|$  are continuous at  $x = 0$ , hence the composite

$$(g \circ f)(x) = \left| \frac{x \sin x}{x^2 + 2} \right| \text{ will also be continuous at } x = 0$$

**Illustration 14**

If  $f(x) = \frac{x+1}{x-1}$  and  $g(x) = \frac{1}{x-2}$ , then discuss the continuity of  $f(x)$ ,  $g(x)$  and  $f \circ g(x)$  in its domain.

**Solution:**

$$f(x) = \frac{x+1}{x-1}$$

$f(x)$  is a rational function it must be continuous in its domain and  $f$  is not defined at  $x = 1$ .

$$g(x) = \frac{1}{x-2}$$

$g(x)$  is also a rational function. It must be continuous in its domain and  $g$  is not defined at  $x = 2$ .

Consider  $g(x) = 1$

$$\frac{1}{x-2} = 1 \Rightarrow x = 3$$

$\therefore f \circ g(x)$  is continuous in its domain :  $\mathbb{R} - \{2, 3\}$

**Illustration 15**

If  $f(x) = \begin{cases} x+a, & x \leq 0 \\ |x-4|, & x > 0 \end{cases}$  and  $g(x) = \begin{cases} x+1, & x < 0 \\ (x-4)^2 + b, & x \geq 0 \end{cases}$  are continuous on  $R$ , then  $(g \circ f)(2) + (f \circ g)(-2)$  is

equal to:

(A) -10

(B) 10

(C) 8

(D) -8

**Ans. (D)**

**Solution:**

$$f(x) = \begin{cases} x+a & ; x \leq 0 \\ |x-4| & ; x > 0 \end{cases} ; g(x) = \begin{cases} x+1 & ; x < 0 \\ (x-4)^2 + b & ; x \geq 0 \end{cases}$$

For continuity  $a = 4$  and  $b = -15$

$$g(f(2)) + f(g(-2))$$

$$= g(2) + f(-1) = -8$$

**5. CONTINUITY OVER COUNTABLE SET:**

There are functions which are continuous over a countable set and else where discontinuous.

**Illustration 16**

If  $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q} \end{cases}$ , find the points where  $f(x)$  is continuous

**Solution:**

Let  $x = a$  be the point at which  $f(x)$  is continuous.

$$\Rightarrow \lim_{\substack{x \rightarrow a \\ \text{through rational}}} f(x) = \lim_{\substack{x \rightarrow a \\ \text{through irrational}}} f(x)$$

$$\Rightarrow a = -a$$

$$\Rightarrow a = 0 \Rightarrow \text{function is continuous at } x = 0.$$

**Illustration 17**

If the function  $f(x) = \begin{cases} \frac{1}{x} \log_e \left( \frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right) & , x < 0 \\ k & , x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1} - 1} & , x > 0 \end{cases}$  is continuous at  $x = 0$ , then  $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$  is equal to:

- (A) -5 (B) 5 (C) -4 (D) 4

**Ans. (A)**

**Solution:**

If  $f(x)$  is continuous at  $x = 0$ ,  $RHL = LHL = f(0)$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1} - 1} \cdot \frac{\sqrt{x^2+1} + 1}{\sqrt{x^2+1} + 1} \quad (\text{Rationalisation})$$

$$\lim_{x \rightarrow 0^+} -\frac{2\sin^2 x}{x^2} \cdot (\sqrt{x^2+1} + 1) = -4$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} \ln \left( \frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right)$$

$$\lim_{x \rightarrow 0^-} \frac{\ln \left( 1 + \frac{x}{a} \right)}{\left( \frac{x}{a} \right)} + \frac{\ln \left( 1 - \frac{x}{b} \right)}{\left( -\frac{x}{b} \right)}$$

$$= \frac{1}{a} + \frac{1}{b}$$

$$\text{So } \frac{1}{a} + \frac{1}{b} = -4 = k$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{4}{k} = -4 - 1 = -5$$

**Illustration 18**

If for  $p \neq q \neq 0$ , then function  $f(x) = \frac{\sqrt[7]{p(729+x)} - 3}{\sqrt[3]{729+qx} - 9}$  is continuous at  $x = 0$ , then:

- (A)  $7pq f(0) - 1 = 0$  (B)  $63q f(0) - p^2 = 0$   
(C)  $21q f(0) - p^2 = 0$  (D)  $7pq f(0) - 9 = 0$

**Ans. (B)**

**Solution:**

$$f(0) = \lim_{x \rightarrow 0} f(x)$$

Limit should be  $\frac{0}{0}$  form

$$\text{So, } \sqrt[7]{p \cdot 729} - 3 = 0 \Rightarrow p \cdot 3^6 = 3^7 \Rightarrow p = 3$$

$$\text{Now, } f(0) = \lim_{x \rightarrow 0} \frac{\sqrt[7]{3(3^6 + x)} - 3}{\sqrt[3]{3^6 + qx} - 9}$$

$$= \lim_{x \rightarrow 0} \frac{3 \left[ \left( 1 + \frac{x}{3^6} \right)^{1/7} - 1 \right]}{9 \left[ \left( 1 + \frac{qx}{3^6} \right)^{1/3} - 1 \right]} = \frac{3}{9} \times \frac{\frac{1}{7 \cdot 3^6}}{\frac{q}{3 \cdot 3^6}}$$

$$\Rightarrow f(0) = \frac{1}{3} \times \frac{3}{7q} = \frac{1}{7q}$$

$$\Rightarrow 7qf(0) - 1 = 0$$

$$\Rightarrow 7 \cdot p^2 \cdot qf(0) - p^2 = 0 \text{ (for option)}$$

$$\Rightarrow 63qf(0) - p^2 = 0$$

**Illustration 19**

Let  $f : [-1, 3] \rightarrow \mathbb{R}$  be defined as  $f(x) = \begin{cases} |x| + [x] & , -1 \leq x < 1 \\ x + |x| & , 1 \leq x < 2 \\ x + [x] & , 2 \leq x \leq 3 \end{cases}$ , where  $[t]$  denotes the greatest integer less

than or equal to  $t$ . Then,  $f$  is discontinuous at:

(A) four or more points

(B) only one point

(C) only two points

(D) only three points

**Ans. (D)**

**Solution:**

$$f(x) = \begin{cases} -(x+1) & , -1 \leq x < 0 \\ x & , 0 \leq x < 1 \\ 2x & , 1 \leq x < 2 \\ x+2 & , 2 \leq x < 3 \\ x+3 & , x = 3 \end{cases}$$

function discontinuous at  $x = 0, 1, 3$

**Illustration 20**

Let  $[t]$  denote the greatest integer  $\leq t$ . The number of points where the function

$$f(x) = [x] |x^2 - 1| + \sin \left( \frac{\pi}{[x] + 3} \right) - [x + 1], x \in (-2, 2) \text{ is not continuous is } \underline{\hspace{2cm}}.$$

**Ans. (2)**

**Solution:**

$$f(x) = [x] |x^2 - 1| + \sin \frac{\pi}{[x] + 3} - [x + 1]$$

$$f(x) = \begin{cases} 3 - 2x^2, & -2 < x < -1 \\ x^2, & -1 \leq x < 0 \\ \frac{\sqrt{3}}{2} + 1 & 0 \leq x < 1 \\ x^2 + 1 + \frac{1}{\sqrt{2}}, & 1 \leq x < 2 \end{cases}$$

discontinuous at  $x = 0, 1$

**Illustration 21**

Let  $\alpha \in \mathbb{R}$  be such that the function  $f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$  is continuous at  $x = 0$ , where

$\{x\} = x - [x]$ ,  $[x]$  is the greatest integer less than or equal to  $x$ . Then:

- (A)  $\alpha = \frac{\pi}{\sqrt{2}}$  (B)  $\alpha = 0$  (C) no such  $\alpha$  exists (D)  $\alpha = \frac{\pi}{4}$

**Ans (C)**

**Solution:**

$$\lim_{x \rightarrow 0^+} f(x) = f(0) = \lim_{x \rightarrow 0^-} f(x)$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2) \cdot \sin^{-1}(1-x)}{x(1-x)(1+x)}$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2)}{x \cdot 1 \cdot 1} \cdot \frac{\pi}{2}$$

$$\text{Let } 1 - x^2 = \cos \theta$$

$$\frac{\pi}{2} \lim_{x \rightarrow 0^+} \frac{\theta}{\sqrt{1 - \cos \theta}}$$

$$\frac{\pi}{2} \lim_{\theta \rightarrow 0^+} \frac{\theta}{\sqrt{2} \sin \frac{\theta}{2}} = \frac{\pi}{\sqrt{2}}$$

$$\text{Now, } \lim_{x \rightarrow 0^-} \frac{\cos^{-1}(1-(1+x)^2) \sin^{-1}(-x)}{(1+x)-(1+x)^3}$$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2}(-\sin^{-1} x)}{(1+x)(2+x)(-x)}$$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2} \cdot \frac{\sin^{-1} x}{x}}{1 \cdot 2} = \frac{\pi}{4}$$

$\Rightarrow \text{RHL} \neq \text{LHL}$

Function can't be continuous

$\Rightarrow$  No value of  $\alpha$  exist

**Illustration 22**

If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a function defined by  $f(x) = [x-1] \cos\left(\frac{2x-1}{2}\right)\pi$ , where  $[.]$  denotes the greatest integer

function, then  $f$  is:

- (A) discontinuous at all integral values of  $x$  except at  $x = 1$   
 (B) continuous only at  $x = 1$   
 (C) continuous for every real  $x$   
 (D) discontinuous only at  $x = 1$

**Ans. (C)**

## Continuity

### Solution:

For  $x = n, n \in \mathbb{Z}$

$$\text{LHL} = \lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^-} [x-1] \cos\left(\frac{2x-1}{2}\right) \pi$$

$$= 0$$

$$\text{RHL} = \lim_{x \rightarrow n^+} f(x) = \lim_{x \rightarrow n^+} [x-1] \cos\left(\frac{2x-1}{2}\right) \pi$$

$$= 0$$

$$f(n) = 0$$

$$\Rightarrow \text{LHL} = \text{RHL} = f(n)$$

$\Rightarrow f(x)$  is continuous for every real  $x$

### Illustration 23

The function  $f(x) = [x]^2 - [x^2]$  (where  $[y]$  is the greatest integer less than or equal to  $y$ ), is discontinuous

at :

(A) all integers

(B) all integers except 0 and 1

(C) all integers except 0

(D) all integers except 1

**Ans. (C)**

### Solution:

$$f(x) = [x]^2 - [x^2]$$

At  $x = I$  (Integer)

$$f(I) = [I]^2 - [I^2] = I^2 - I^2 = 0$$

$$f(I^+) = \lim_{h \rightarrow 0} [I+h]^2 - [(I+h)^2]$$

$$= I^2 - I^2 = 0$$

$$f(I^-) = \lim_{h \rightarrow 0} [I-h]^2 - [(I-h)^2]$$

$$= (I-1)^2 - (I^2 - 1) = I^2 + 1 - 2I - I^2 + 1$$

$$= 2 - 2I$$

$\Rightarrow f(x)$  is discontinuous at all integers & except 0.

$\therefore$  At  $x = 1$ ,

$$f(1) = f(1^+) = f(1^-) = 0$$

and At  $x = 0$ ,

$$f(0) = 0 = f(0^+)$$

$$f(0^-) = 1 =$$